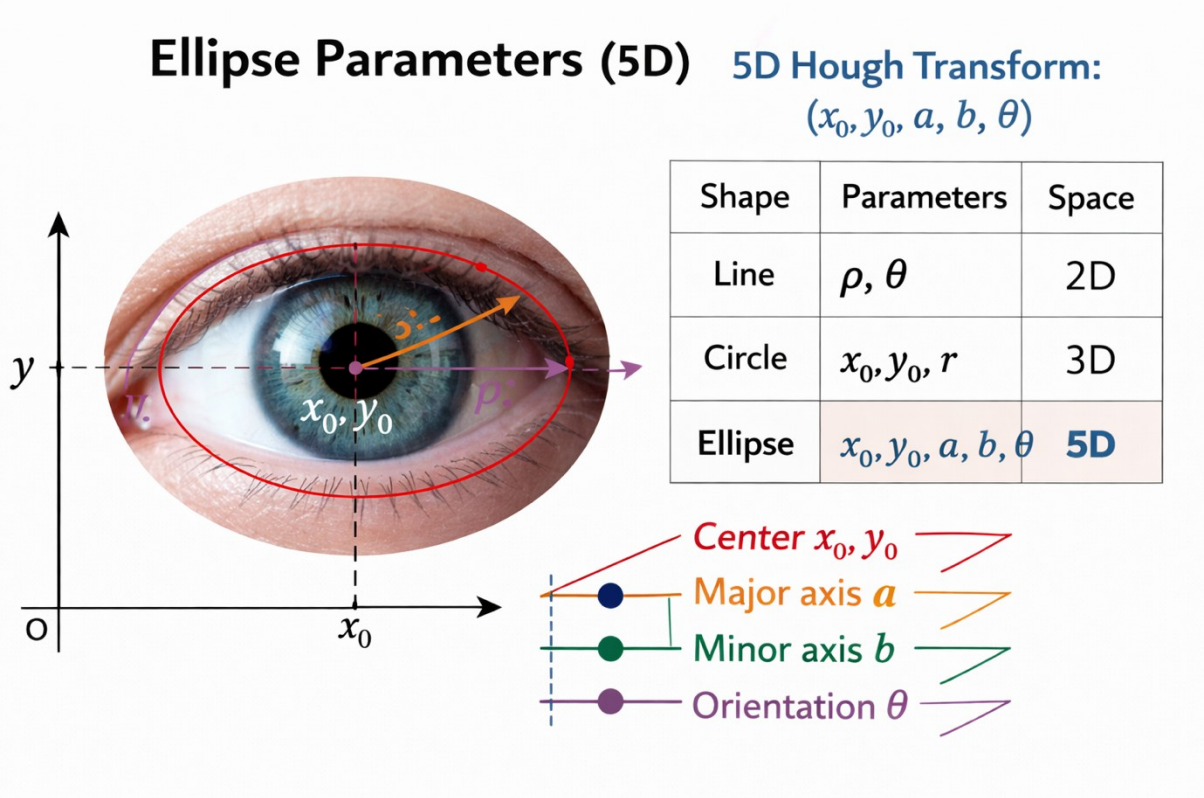
Unit 3 - Session 5

SLO 1: Analyze the 5D parameter space requirements for ellipse detection

Duration: 10 Minutes Activity

Question:



1. What parameters are required to represent an ellipse in the Hough Transform?

Answer: An ellipse requires five parameters: center (x_0, y_0), major axis (a), minor axis (b), and orientation (θ).

2. Why does ellipse detection require a higher-dimensional parameter space compared to line and circle detection?

Answer: An ellipse has five independent parameters, compared with two for a line and three for a circle, so it needs a higher-dimensional parameter space.

3. How does each parameter (x_0, y_0, a, b, θ) relate to the geometry of an ellipse?

Answer: x_0, y_0 define the center; a defines the major axis; b defines the minor axis; θ defines the orientation of the ellipse.

4. What practical challenges arise due to the 5D parameter space in ellipse detection?

Answer: The 5D parameter space requires a large accumulator, leading to high memory usage and increased computational time.

5. Why is ellipse detection suitable for modeling the human iris instead of circle detection?

Answer: The iris may appear elliptical because of the viewing angle and perspective, so an ellipse models its shape more accurately than a circle.